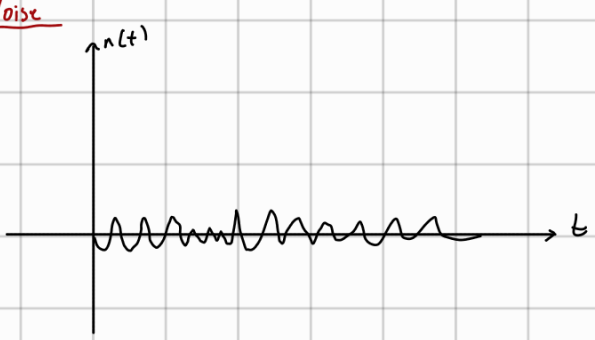
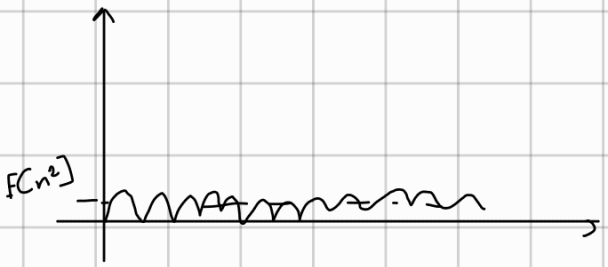
Noise

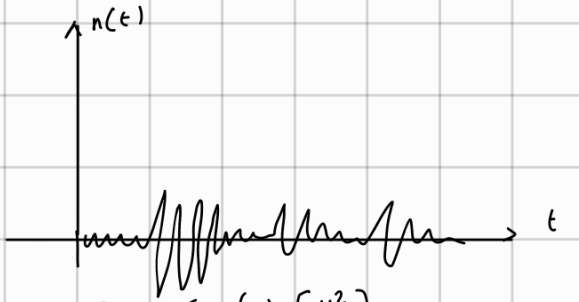


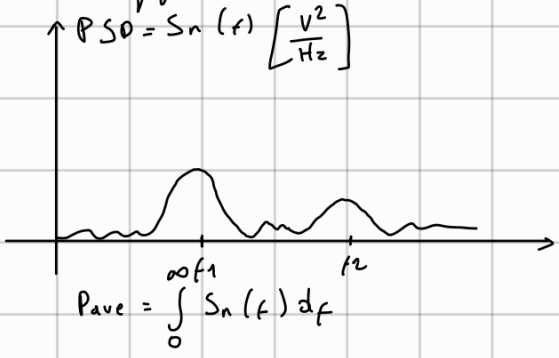


$$P_{ave} = \lim_{T\to\infty} \int_0^T n^2(t)\,dt = E[n^2(t)]$$

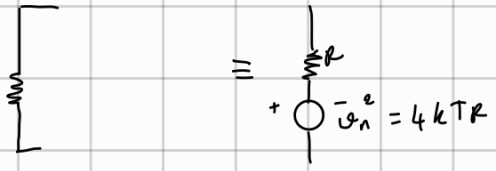
$$\underbrace{\sigma^2}_{P_{AC}} = \underbrace{E[n^2]}_{P_{ave}} - \underbrace{E^2[n]}_{P_{DC}}$$

$$P_{ave} = P_{AC} + P_{DC}$$

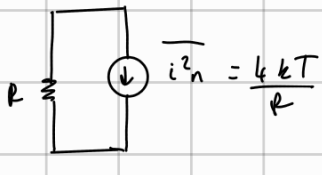




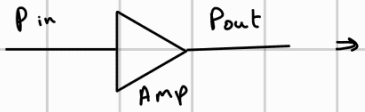
$$P_{ave} = \int_0^{\infty} S_n(f)\,df$$



or

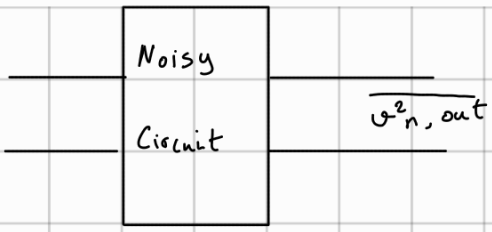


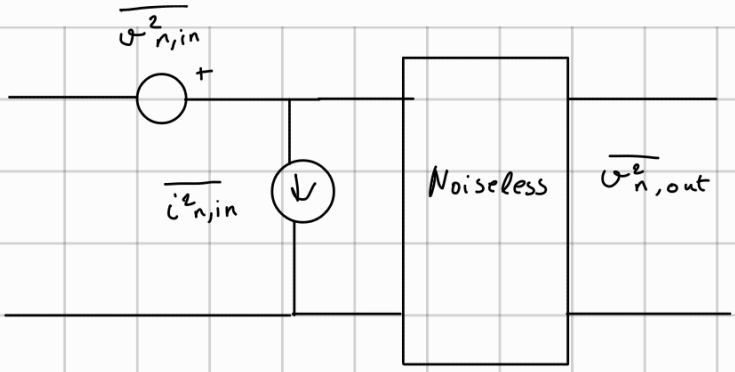
Input referred Noise



$$P_{out} = P_{in} \cdot G$$

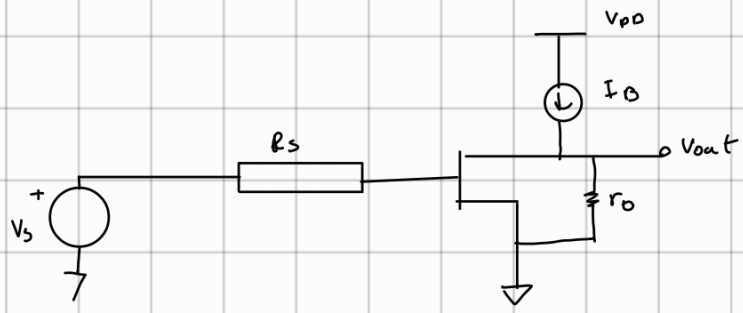
$$N_{out} = N_{in} \cdot G + N_A$$





How to find input referred Noise

Ex



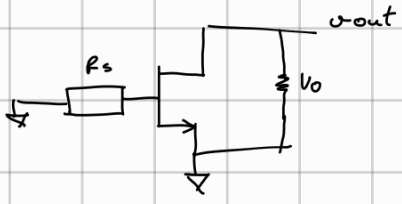
$$\overline{v_{n,out1}^2} + \overline{v_{n,out2}^2} + \overline{v_{n,out3}^2} + \cdots = \overline{v_{n,out\,(total)}^2}$$

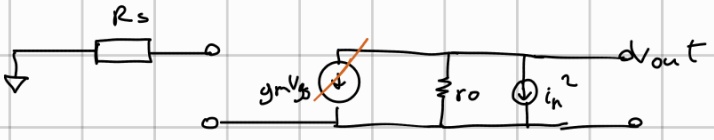
$$\overline{v_{n,input}^2} = \frac{\overline{v_{n,out\,(total)}^2}}{A_{vt}^2}$$

① Cancel sources (gerilim kısa devre, akım açık)

$v_s \rightarrow$ short

$I_B \rightarrow$ open





$$\overline{i_n^2} = 4kT\gamma g_m$$

$$\Rightarrow \overline{v_{out,1}^2} = \overline{i_n^2} \cdot r_o^2$$

$$\overline{v_{out,1}^2} = r_o^2\, 4kT\gamma g_m$$

②



Transistor gain

$$\overline{v_{n,out2}^2} = \overline{v_{n,R_S}^2}\,(g_m r_o)^2 = 4kTR_S\,(g_m r_o)^2$$

③ Sum of squares

$$\overline{v_{n,out\,(total)}^2} = \overline{v_{n,out1}^2} + \overline{v_{n,out2}^2} = r_o^2\, 4kT\gamma g_m + (g_m r_o)^2\, 4kTR_S$$

(4) $\overline{v_{in,n}^2} = \dfrac{\overline{v_{n,out\ (total)}}}{A_{vt}^2} = \dfrac{r_o^2\, 4kT\gamma g_m + (g_m r_o)^2\, 4kTR_s}{(g_m r_o)^2}$

$$\boxed{\overline{v_{in,n}^2} = \frac{4kT\gamma}{g_m} + 4kTR_s}$$

$\frac{4kT\gamma}{g_m}$ → amplifierden gelen gürültü

$4kTR_s$ → kaynağın gürültüsü

to decrease $\overline{v_{in,n}^2}$, increase g_m

genellikle amplifierlar 10, 15 dB üstü tasarlanır ki gürültüsü az olsun.

Noise Figure (NF) < dB

Noise Factor (F)

$$\boxed{NF = 10 \log(F)}$$



$$SNR_{in} = \frac{P_{s,in}}{P_{n,in}}$$

$$SNR_{out} = \frac{P_{s,out}}{P_{n\,out}} = \frac{P_{s,out}}{P_{n,in} \cdot G + P_{n,A}}$$

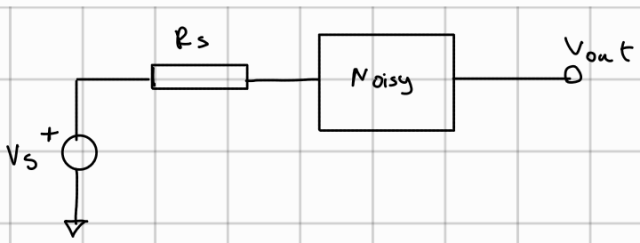
$SNR_{in} > SNR_{out}$

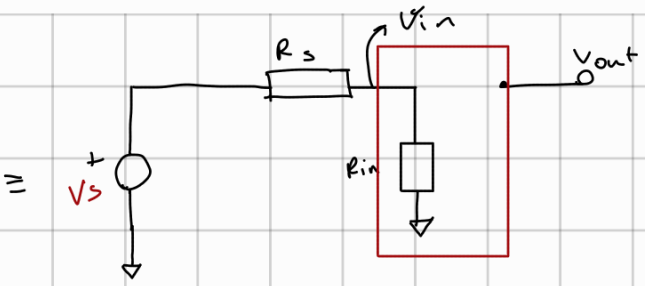
$$\boxed{F = \frac{SNR_{in}}{SNR_{out}}} > 1$$

$$\boxed{NF = 10 \log \underbrace{\left[\frac{SNR_{in}}{SNR_{out}}\right]}_{F}}$$

$$NF = SNR_{in_{dB}} - SNR_{out_{dB}} > 0\,dB$$

A good LNA → NF: 0 ~ 2 dB



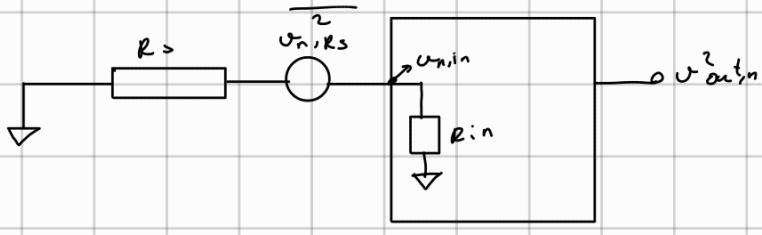


$$SNR_{in} = \frac{P_{s,in}}{P_{n,in}}$$

$$P_{s,in} \rightarrow v_{in} = \frac{R_{in}}{R_{in}+R_s}\, v_s$$

$$P_{s,in} = \frac{v_{in}^2}{R_{in}} = \frac{\left(\frac{R_{in}}{R_{in}+R_s}\right)^2 v_s^2}{R_{in}}$$

$P_{n,in} \rightarrow$



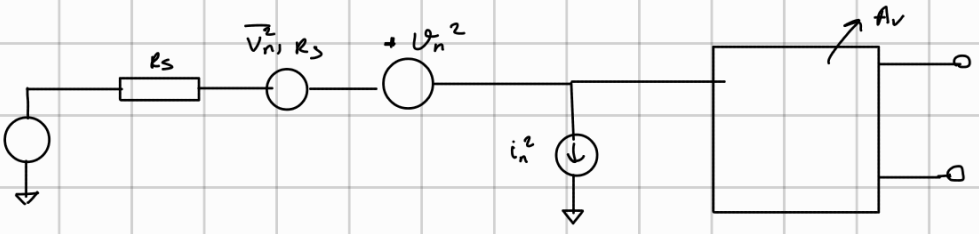
$$\rightarrow v_{n,in} = \frac{R_{in}}{R_{in}+R_s} \cdot v_{n,Rs}$$

$$P_{n,in} = \frac{\overline{v_{n,in}^2}}{R_{in}} = \frac{\left(\frac{R_{in}}{R_{in}+R_s}\right)^2 \overline{v_{n,Rs}^2}}{R_{in}}$$

$$SNR_{in} = \frac{P_{s,in}}{P_{n,in}} = \frac{\left\{\frac{\left(\frac{R_{in}}{R_{in}+R_s}\right)^2 \cdot v_s^2}{R_{in}}\right\}}{\left\{\frac{\left(\frac{R_{in}}{R_{in}+R_s}\right) \overline{v_{n,Rs}^2}}{R_{in}}\right\}} = \frac{v_s^2}{\overline{v_{n,Rs}^2}}$$ ← sadece kaynağa bağlı

$$SNR_{in} = \frac{v_s^2}{\overline{v_{n,Rs}^2}}$$

$$SNR_{out} = \frac{P_{s,out}}{P_{n,out}}$$

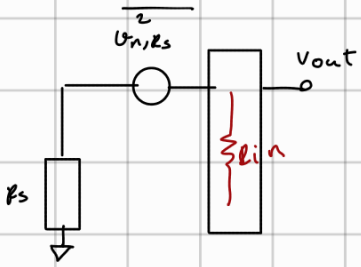


$$P_{s,out} = \left[V_s \left(\frac{R_{in}}{R_{in}+R_s}\right)\right]^2 \cdot A_v^2 = V_s^2 \left(\frac{R_{in}}{R_{in}+R_s}\right)^2 \cdot A_v^2$$

$P_{n,out}$

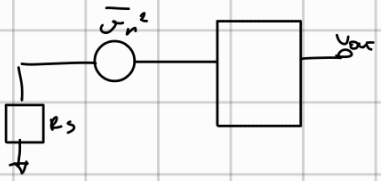
↳ ① Noise coming from R_S ($\overline{v_n^2}=0$, $\overline{i_n^2}\to\infty$)

$$\overline{v_{n,out_n}^2} = \overline{v_{n,R_S}^2} \cdot \left(\frac{R_{in}}{R_{in}+R_S}\right)^2 \cdot A_v^2$$



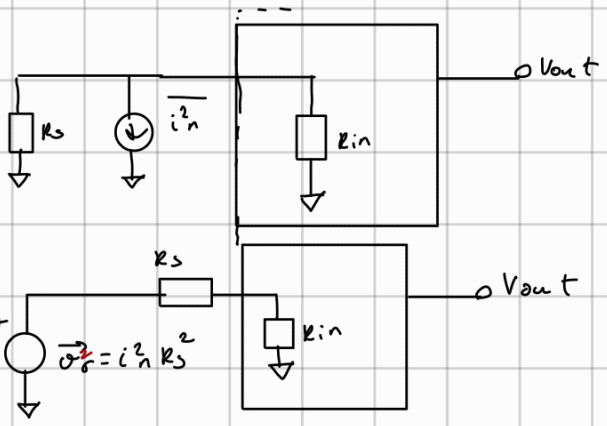
② Noise coming from $\overline{v_n^2}$ ($\overline{v_{n,R_S}}=0$, $\overline{i_n^2}\to\infty$)

$$\overline{v_{n,out_2}^2} = \overline{v_n^2} \left(\frac{R_{in}}{R_{in}+R_S}\right)^2 \cdot A_v^2$$



③ Noise coming from i_n^2 ($\overline{v_n^2}=0$, $\overline{v_{n,R_S}^2}=0$)

$$\overline{v_{n,out_3}^2} = i_n^2 R_S^2 \left(\frac{R_{in}}{R_{in}+R_S}\right)^2 \cdot A_v^2$$



$$SNR_{out} = \frac{P_{s,out}}{P_{n,out}} = \frac{V_s^2 \left(\frac{R_{in}}{R_{in}+R_S}\right)^2 \cdot A_v^2}{\left[\overline{v_{n,R_S}^2} + \overline{v_n^2} + \overline{i_n^2} R_S^2\right]\left(\frac{R_{in}}{R_{in}+R_S}\right)^2 A_v^2}$$

$$\boxed{SNR_{out} = \frac{V_s^2}{\overline{v_{n,R_S}^2} + \overline{v_n^2} + \overline{i_n^2} \cdot R_S^2}}$$

$$F = \frac{SNR_{in}}{SNR_{out}} = \frac{V_s^2 / \overline{v_{n,R_S}^2}}{V_s^2 / \left(\overline{v_{n,R_S}^2} + \overline{v_n^2} + \overline{i_n^2} \cdot R_S^2\right)}$$

$$F = \frac{\overline{v_{n,R_S}^2} + \overline{v_n^2} + \overline{i_n^2} R_S^2}{\overline{v_{n,R_S}^2}}$$

$$\boxed{F = 1 + \frac{\overline{v_n^2} + \overline{i_n^2} R_S^2}{\overline{v_{n,R_S}^2}}}$$

→ input referred noise

→ source noise

$$A_v = \frac{v_{out}}{v_{in}}$$

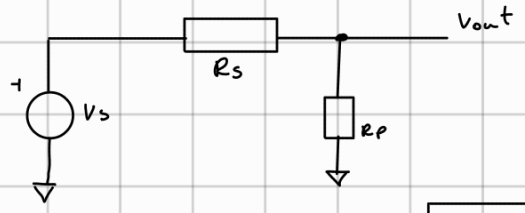
$$A_{v_t} = \frac{v_{out}}{v_s} = \frac{v_{out}}{v_{in}} \cdot \frac{v_{in}}{v_s}$$

$$F = \frac{A_{vt}^2}{A_{vt}^2} \left(\frac{\overline{v_{n,R_S}^2} + \overline{v_n^2} + \overline{i_n^2} \cdot R_S^2}{\overline{v_{n,R_S}^2}}\right)$$

$$= \frac{\overline{v_{n,out}^2}}{A_{vt}^2} \cdot \frac{1}{\overline{v_{n,R_S}^2}}$$

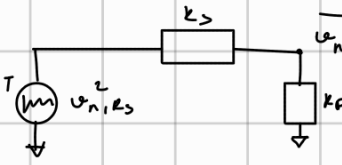
$$\Rightarrow \boxed{F = \frac{\overline{v_{n,out}^2}}{A_{vt}^2} \cdot \frac{1}{4kTR_S}}$$

Ex:



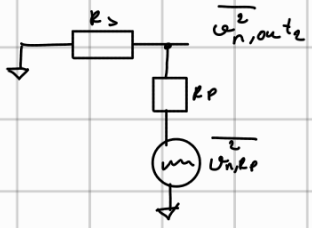
$$F = \frac{\overline{v_{n,out}^2}}{A_{vt}^2} \cdot \frac{1}{4kTR_s}, \qquad \boxed{A_{vt} = \frac{R_p}{R_s+R_p}}$$

$\overline{v_{n,out}^2}$ (total) → R_s ($V_s=0$) →



$$\overline{v_{n,out_1}^2} = \left(\frac{R_p}{R_p+R_s}\right)^2 \overline{V_{n,R_s}^2}$$

R_p ($V_s=0$) →



$$\overline{v_{n,out_2}^2} = \left(\frac{R_s}{R_s+R_p}\right)^2 \cdot \overline{V_{n,R_p}^2}$$

$$\overline{v_{n,out\,(total)}^2} = \left(\frac{R_s}{R_p+R_s}\right)^2 \cdot \underbrace{\frac{\overline{v_{n,R_p}^2}}{4kTR_p}} + \left(\frac{R_p}{R_p+R_s}\right)^2 \cdot \underbrace{\frac{\overline{v_{n,R_s}^2}}{4kTR_s}}$$

$$= 4kT\,\frac{R_sR_p^2 + R_pR_s^2}{(R_p+R_s)^2} = 4kT\,\frac{R_sR_p(R_s+R_p)}{(R_s+R_p)^2}$$

$$\boxed{\overline{v_{n,out\,(total)}^2} = 4kT\,\underbrace{\frac{R_sR_p}{R_s+R_p}}_{R_s//R_p}}$$

$$F = \frac{\overline{v_{n,out}^2}}{\dfrac{4kT\left(\frac{R_sR_p}{R_s+R_p}\right)}{\frac{R_p^2}{(R_s+R_p)^2}}} \cdot \frac{1}{4kTR_s}$$

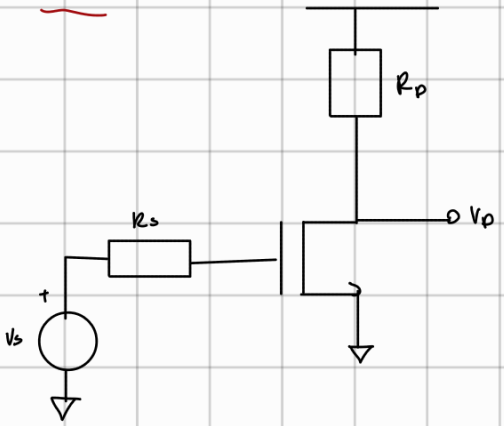
($\overline{v_{n,out}^2}$; A^2v_t)

$$\boxed{F = \frac{R_s+R_p}{R_p} = 1+\frac{R_s}{R_p}}$$

$R_p \uparrow\uparrow \longrightarrow F \approx 1$ görültüsüz

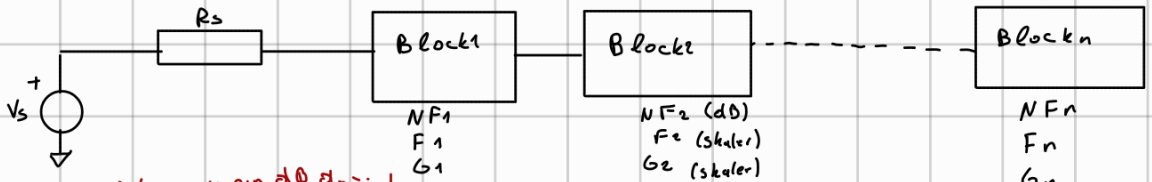
$R_p \gg R_s \longrightarrow$ No noise effect coming from R_p

Ex:



$$F = \frac{v_{n,out}^2}{A^2 v_t} \qquad \frac{1}{4kTR_s} = ?$$

Noise in Cascaded Stages



↘ skaler işlem yapmak için dB değişir

Friis' Formula

$$F(total) = 1 + (F_1 - 1) + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \frac{F_4 - 1}{G_1 G_2 G_3} + \cdots + \frac{F_n - 1}{G_1 G_2 \cdots G_{n-1}}$$

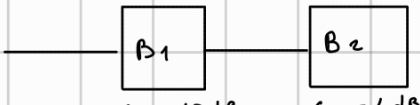
$$F(total) = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2}$$

$\frac{F_2 - 1}{G_1} \approx 10dB$

Dominant F

mutlaka girişte düşük gürültülü ve NF düşük olan amplifier seçilmeli

Ex



$G_1 = 10dB$ $\quad G_2 = 4dB$

$NF_1 = 2dB$ $\quad NF_2 = 10dB$

$NF_t = ?$

$$F_t = F_1 + \frac{F_2 - 1}{G_1} = 1,58 + \frac{10-1}{10} = 1,58 + 0,9 = 2,48$$

$$F_1 = 10^{\frac{NF_1}{10}} = 1,58$$

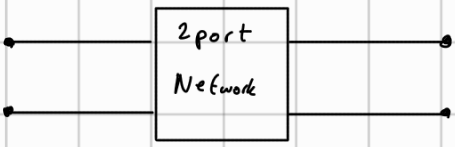
$$F_2 = 10^{\frac{NF_2}{10}} = 10$$

$$G_1 = 10^{\frac{10}{10}} = 10$$

$$\longrightarrow NF_t = 10 \log F_t = 10 \log (2,48) = 3,95dB$$

Noise in Passive Reciprocal Circuit

Reciprocity:



$$S = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}$$

S_{11} → Return loss

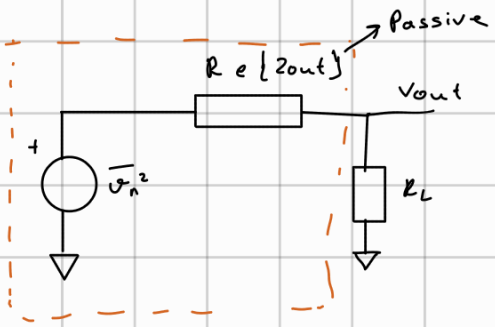
S_{22} → Output Return loss

S_{21} → Gain

S_{12} → Isolation

Reciprocity ← pasif devrelerde olabiliyor

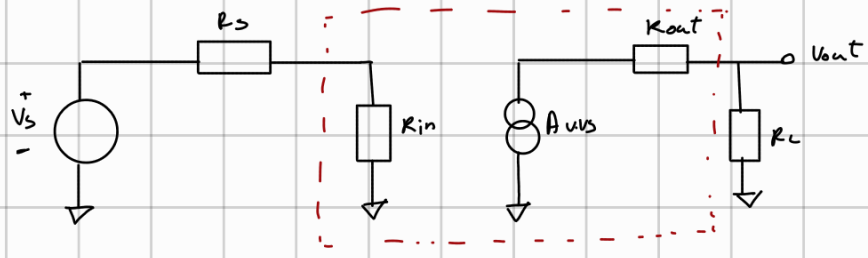
$S_{12} = S_{21}$



$$\overline{v_n^2} = 4kT\,\underbrace{Re\{Zout\}}_{Rout}$$

$$\overline{v_{n,out}^2} = \overline{v_n^2}\left(\frac{R_L}{R_L + Rout}\right)^2$$

$$\overline{v_{n,out}^2} = 4kTR_{out}\left(\frac{R_L}{R_L + Rout}\right)^2$$



$R_s = R_{in}$

$\frac{R_{in}}{R_s + R_{in}} = \frac{1}{2}$

$R_L = R_{out}$

Input matched

output matched

$$P_{in} = \frac{V_{in}^2}{R_{in}} = \frac{V_s^2}{4R_{in}}$$

$$v_{in} = \frac{1}{2}V_s$$

$$v_{in}^2 = \frac{v_s^2}{4}$$

$$P_{out} = \frac{(A_v \cdot V_s)^2}{4 R_L}$$

$$\frac{P_{out}}{P_{in}} = G$$

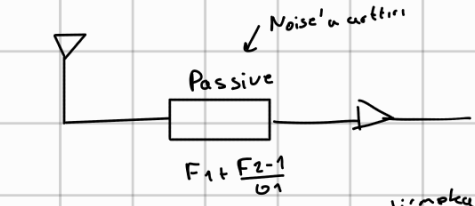
$$G = \frac{\left\{\frac{(A_v \cdot V_s)^2}{4 R_L}\right\}_{P_{out}}}{\left\{\frac{V_s^2}{4 R_{in}}\right\}_{P_{in}}} = \boxed{A_v^2 \cdot \frac{R_s}{R_L} = G}$$

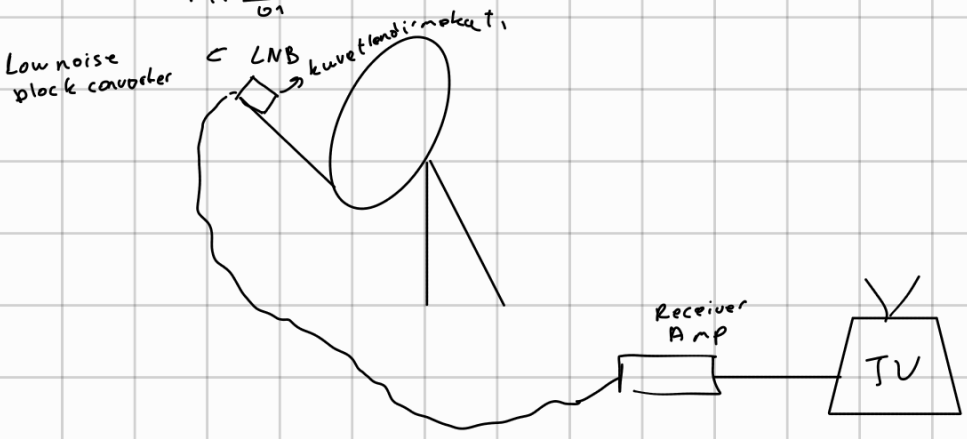
$$\boxed{\underbrace{L}_{\text{zayıflatma}} = \frac{1}{G} = \frac{1}{A_v^2} \cdot \frac{R_L}{R_s}}$$

Pasif devrelerde $\Rightarrow$ $\Leftarrow$ //

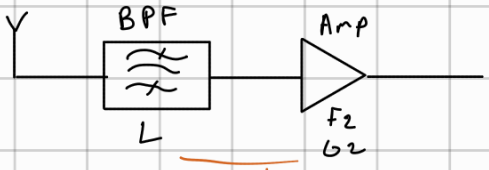
$$\cancel{NF} = \frac{\bar{v}_{n,out}^2}{A_{vt}^2} \cdot \frac{1}{4kTR_s} = \frac{\frac{4kTR_L}{4}}{\left(\frac{A_v}{2}\right)^2} \cdot \frac{1}{4kTR_s} \Rightarrow \boxed{\cancel{NF} = \frac{1}{A_v^2} \cdot \frac{R_L}{R_s}}$$

$$A_{vt} = \left(\frac{V_o}{V_{in}}\right)^{A_v} \left(\frac{V_{in}}{V_s}\right)^{1/2} = \frac{A_v}{2}$$





Ex:



$F_t = ?$

$$F_t = F_1 + \frac{F_2 - 1}{G_1} = L + \frac{F_2 - 1}{\frac{1}{L}} = L + LF_2 - L = LF_2 = F_{total}$$

$F_1 = L$

$G_1 = \frac{1}{L}$

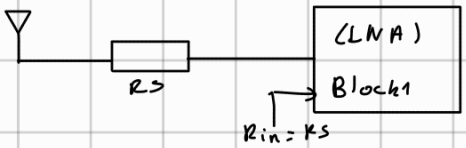
$$\boxed{NF_t = L_{dB} + NF_2}$$

Noise figürü arttırı

ters olmasıdahaiyi

Onun gerine dar bantlı amplisey

Noise power in Rx input



$$\overline{v_{in,n}^2} = \left(\frac{R_{in}}{R_{in}+R_s}\right)^2 \cdot \overline{v_s^2} = kTR_s$$

$\left(\frac{R_{in}}{R_{in}+R_s}\right) = \frac{1}{4}$, $\overline{v_s^2} \downarrow 4kTR_s$

$$\frac{\overline{v_{n,in}^2}}{R_{in}} = \frac{kTR_s}{R_{in}} = kT \quad (1Hz\ BW)$$

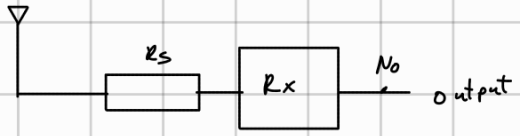
$R_{in} = R_s$

1Hz'lik banttaki kT kadar power bulunur

$P_{in} = kTB$ (Input noise power in bandwith B)

Receive Noise Floor

Noise floor = Receiver input equivalent noise



N_o : output noise power exluding R_s.

kaynak düşünülmediğindeki gürültü

$$F = \frac{\overline{v_{n,out}^2}}{A_v^2} \cdot \frac{1}{4kTR_s} = \frac{\text{Total output noise power including } R_s}{G \cdot (\text{Noise power from } R_s \text{ only})} = \frac{N_o + G(kTB)}{G \cdot kTB} = F$$

$$\Rightarrow \text{Input noise} = kTB + \frac{N_o}{G} = \underbrace{FkTB}_{\text{Noise floor}}$$